AES (Advanced Encryption Standard)

1. Overview

AES is a **symmetric-key block cipher** established as a specification for encrypting electronic data by the U.S. **National Institute of Standards and Technology (NIST) in 2001**, replacing the aging DES standard.

It is the most widely used encryption algorithm in the world today, underpinning everything from disk encryption to VPNs to TLS.

AES at a Glance

Property	Detail
Type	Symmetric-key block cipher
Block size	Fixed at 128 bits
Key sizes	128, 192, or 256 bits
Structure	Substitution-Permutation Network (SPN) — **not** a Feistel network
Standardized by	NIST, 2001 (based on the Rijndael algorithm)

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2. Why AES Replaced DES

DES's **56-bit key length** became increasingly vulnerable to brute-force attacks as computing power grew. NIST ran an open, multi-year public competition; the **Rijndael algorithm**, designed by **Joan Daemen and Vincent Rijmen**, was selected as the winner and standardized as AES.

Key Advantages Over DES and 3DES

- **Much larger key space** — 128/192/256-bit vs. DES's 56-bit.
- **Symmetric, parallelizable structure** — well suited to modern processor architectures, unlike DES's inherently sequential Feistel rounds.
- **No publicly known practical attack against the full-round cipher** when used correctly.

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3. Internal Structure

AES operates on a **4×4 grid of bytes** called the **state**, derived from the 128-bit input block.

Encryption proceeds through a number of rounds determined by the key size:

Key Size	Number of Rounds
128-bit	10
192-bit	12
256-bit	14

AES Round Transformations

Each round, except the last, applies four transformations to the state. The final round omits **MixColumns**.

1. **SubBytes** — a non-linear byte substitution using a fixed S-box, providing confusion.
2. **ShiftRows** — a transposition step that cyclically shifts each row of the state by an offset, providing diffusion across columns.
3. **MixColumns** — a linear mixing operation treating each column as a polynomial, combining bytes within a column. This is omitted in the final round.
4. **AddRoundKey** — a bitwise XOR of the state with a round key derived from the main key via the **key schedule**.

An initial `AddRoundKey` step is applied before the first round begins.

Simplified AES Flow

```
128-bit Plaintext Block
          |
          ▼
      AES State
       4 × 4 Bytes
          |
          ▼
    Initial AddRoundKey
          |
          ▼
     AES Rounds
          |
          ├── SubBytes
```

```
            ├── ShiftRows
            ├── MixColumns
            └── AddRoundKey
            |
            ▼
        Final Round
     (No MixColumns)
            |
            ▼
        Ciphertext
```

4. Modes of Operation

AES is a block cipher, meaning it natively only encrypts a single fixed-size block. To encrypt data of arbitrary length, it must be combined with a **mode of operation**.

Mode	Behavior	Notes
ECB (Electronic Codebook)	Each block encrypted independently	**Insecure for most uses** — identical plaintext blocks produce identical ciphertext blocks, leaking patterns.
CBC (Cipher Block Chaining)	Each block XORed with the previous ciphertext block before encryption	Requires an unpredictable IV; sequential, not parallelizable for encryption.
CTR (Counter)	A counter value is encrypted and XORed with plaintext, turning the block cipher into a stream cipher	Fully parallelizable; requires a unique nonce/counter per block.
GCM (Galois/Counter Mode)	CTR mode combined with a built-in authentication tag	Provides both **confidentiality and integrity/authenticity (AEAD)** — the standard choice for modern protocols like TLS.

Mode Selection

```
AES Block Cipher
       |
       ▼
Mode of Operation
       |
       ├── ECB → ❌ Avoid for most uses
       ├── CBC → Requires careful IV handling
```

```
        ├── CTR → Requires unique nonce/counter
        └── GCM → ✅ Confidentiality + Integrity
```

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5. Security Considerations

- AES itself has **no known practical cryptanalytic break against its full round count**; security concerns in real deployments almost always trace back to **mode-of-operation misuse** (e.g., ECB, reused IVs/nonces) or **key management failures**, not the cipher primitive itself.
- **Side-channel attacks** (timing, power analysis, cache-timing) against naive software implementations are a real concern; modern CPUs mitigate this via dedicated **AES-NI hardware instructions** that run in constant time.
- **AES-256** is often chosen for long-term or highly sensitive data, partly as a hedge against future cryptanalytic advances and the eventual arrival of large-scale quantum computers. Quantum computers reduce effective symmetric key strength via **Grover's algorithm**, roughly halving the effective bit-strength, making AES-256 the safer long-term choice over AES-128.

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6. Typical Use Cases

- **Full-disk and file encryption** — BitLocker, VeraCrypt, FileVault
- **VPN tunnel encryption** — IPsec, WireGuard, OpenVPN
- **TLS/SSL bulk data encryption** — via AES-GCM cipher suites
- **Wireless security** — WPA2/WPA3 use AES as the underlying cipher
- **Database and application-level encryption at rest**

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7. Summary

AES is a **fast, well-vetted, hardware-accelerated symmetric block cipher** and the default choice for bulk data encryption today.

Its security in practice hinges almost entirely on correct **mode-of-operation selection** (prefer authenticated modes like GCM) and **sound key management**. The algorithm itself, used correctly, remains unbroken.